Adding and subtracting polynomials

1 Consider the term $9x^8$.

a Identify the coefficient of $9x^8$.

b State the exponent.

2 Consider the expression $5y^3$.

a Determine the number of terms in the expression.

b Identify the coefficient of the term.

3 Consider the expression $-2a^3$.

a State whether this expression is a monomial.

b State the variable, the coefficient, and the degree.

4 For each of the following expressions:

i Determine the number of terms in the expression.

ii Identify the coefficients of the terms.

a $-5y^9 - y$

b $x + 7x^6 - x^9$

5 State whether each of the following is a monomial:

a $2x^3 - 4x^5 + 3$

b $3x^3 + \dfrac{2}{x^7} - 1$

c $\sqrt{2}x^6 + 7x$

6 State whether each of the following statements is true or false.

a The degree of a polynomial in x is the largest coefficient of any of the terms of the polynomial.

b The degree of a polynomial is not always the exponent of the first term.

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7 For each of the following polynomials, identify:

i The degree

ii The leading coefficient

iii The constant term

a $P(x) = 2x^7 + 2x^5 + 2x + 2$

b $P(x) = 6 - \frac{7}{10}x^3$

c $P(x) = \frac{x^7}{5} + \frac{x^6}{6} + 5$

8 Simplify the following:

a $(-4x^2 + 7x - 9) + (6x^2 + 7x + 8)$

b $(-8x^2 - 3) + (-4x^2 - 8x)$

c $(3x^3 - 9x^2 - 8x - 7) + (-7x^3 - 9x)$

d $\left(\frac{1}{2}x^2 - \frac{1}{4}x + \frac{1}{2}\right) + \left(\frac{1}{3}x^2 + \frac{2}{5}x - \frac{1}{4}\right)$

e $(-0.2 - 9.5x - 1.4x^2) + (2.6 - 7.8x + 7.7x^2)$

f $(4x^2y - 9xy^2 + 4) + (-7x^2y + 4xy^2 + 8)$

g $(3x^2y^2 - 2xy^2 + 4y^2) + (-9x^2y^2 + 3xy^2 - 8y^2)$

h $(3x^3 - 3x^2 - 8) + (-5x^3 + 9x^2 + 1)$

i $3(-2x + 7) - (7x^2 + 8x - 5) + 7x$

j $-6x^3 + 4x^2 + 4x - (3x^3 + x + 1)$

k $(-5x^2 + 6x - 4) - (-7x^2 + 7x + 9)$

l $(-2x^3 + 7x^2 + 9x - 2) - (9x^3 + 7x^2)$

m $(4x^2y - 5xy^2 - 4) - (-8x^2y - 6xy^2 - 9)$

n $(-x^3 - 3x^2 - 4) - (x^3 - 9x - 9)$

o $\left(\frac{2}{3}x^2 + \frac{1}{2}x - \frac{2}{3}\right) - \left(\frac{1}{4}x^2 - \frac{2}{3}x + \frac{1}{4}\right)$

p $(7.6x^2 - 0.5x + 7.3) - (1.4x^2 - 3.4x - 9.6)$

q $(x^3 - 4x^2 + 4) + (8x^2 - 2) - (-2x^3 + 4x^2 + 8x + 9)$

r $(-x^2 + 8x - 6) + (4x^2 - x + 7)$

9 If a picture frame has a length of $8x^2 - 9x + 3$ and a width of $6x^3 + 9x^2$, form an expression for the perimeter of the rectangular picture frame.

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10 Nadia's hat manufacturing business sells its hats exclusively through two retailers. The profit generated through selling at Just Stuff is modelled by $A(q) = -0.5q^2 + 22.5q + 510$ and the profit generated through selling at Glorious Gifts is modelled by $B(q) = 2.9q^2 + 36.5q + 70$, where q is the number of hats sold.
Form an expression for the polynomial, $C(q)$, that models Nadia's total profit.

11 The revenue generated by Tricia's seafood restaurant is modeled by $R(m) = -3.2m^2 + 10.9m + 990$ and the profit of her restaurant is modeled by $P(m) = 2.3m^2 - 29.3m + 830$, where m is the number of meals produced.
Find the polynomial that models the costs of Tricia's restaurant.

12 If $P(x) = -x^2 - 3x - 6$ and $Q(x) = 6x - 7$, form a simplified expression for $P(x) + Q(x)$.

13 If $P(x) = -5x^2 - 6x - 6$ and $Q(x) = -7x + 7$, form a simplified expression for $P(x) - Q(x)$.

14 If $A(x) = -5x^2 + 5$, $B(x) = -7x + 6$ and $C(x) = -2x^2 - 2x$, form a simplified expression for $A(x) + B(x) + C(x)$.

15 If $A(x) = 6x^2 - 6$, $B(x) = 3x - 3$ and $C(x) = -2x^2 + 7x$, form a simplified expression for $A(x) - B(x) - C(x)$.

16 If $A(x) = -6x^2 - 6$, $B(x) = 4x - 7$ and $C(x) = x^2 - 5x$, form a simplified expression for $B(x) - A(x)$.

17 Consider the polynomials $P(x) = 7x^2 + 4x - 8$ and $Q(x) = 5x - 7$.

- **a** Determine the degree that $P(x) - Q(x)$ would have.
- **b** State what type of polynomial would $P(x) - Q(x)$ be.
- **c** Solve for $P(x) - Q(x)$.

18 $R(x)$ is the result of adding polynomials $P(x)$ and $Q(x)$, where $P(x) = 3x - 6x^2 + 7$ and $Q(x) = 2x^3 - 4x - 3$.

- **a** Express $R(x)$ in simplest form.
- **b** Determine the degree of $R(x)$.
- **c** Identify the leading coefficient of $R(x)$.
- **d** State whether $R(x)$ is a polynomial.

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19 Consider the the sum of polynomials $P(x) = 4x^4 - 5$ and $Q(x) = 2x^6 + 4x^4 + 3$. Identify:

a The degree of the sum of the polynomial.

b The constant term of the sum of the polynomial.

20 Consider the polynomials $P(x) = 4x^3 + 4$ and $Q(x) = 8 - 4x^3$.

a Write the simplified expression for $P(x) + Q(x)$.

b What is the degree of $P(x) + Q(x)$?

c State whether $P(x) + Q(x)$ is classified as a polynomial.

21 If $P(x) = 3x^3 - 7x^2 - x + 2$, $Q(x) = 3 - 9x^3$ and $R(x) = -3x^2 + 9x - x^3 + 6$, find an expression for each of the following in simplest form:

a $P(x) - R(x)$

b $P(x) - (R(x) - Q(x))$

22 State the degree of the resulting polynomial in each case.

a $P(x) + Q(x)$, where $P(x)$ has degree 4 and $Q(x)$ has degree 6.

b $P(x) - Q(x)$, where $P(x)$ has degree 4 and $Q(x)$ has degree 6.

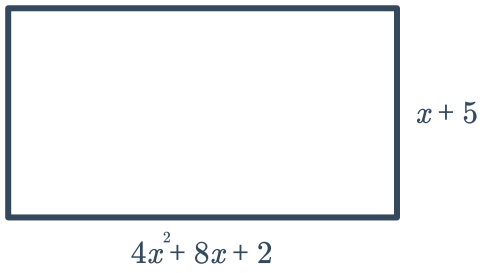
c $P(x) - Q(x)$, where $P(x)$ has degree 6 and $Q(x)$ has degree 6, and the leading coefficient of $P(x)$ is not equal to the leading coefficient of $Q(x)$.

23 Let $P(x) = ax^2 + (b-5)x - 1$ and $Q(x) = x^2 - 5x + 2$.
If $R(x) = 2x^2 + 2x + 1$ is the result of adding $P(x)$ and $Q(x)$, find the values of a and b.

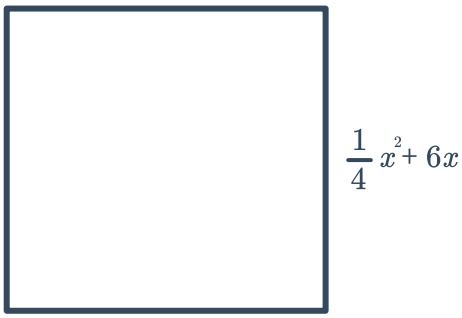
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24 Find a polynomial that represents the perimeter of each of the following polygons.

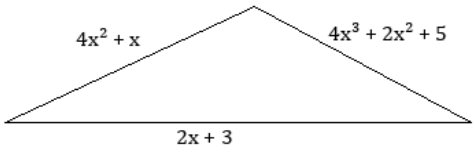
a



b



c



Additional questions

26 Consider the expression $5 + 5x^2 + 13 + \frac{x^2}{5} + \frac{1}{5} + 2x$.

Determine whether each of the following terms could be combined with 5.

a $2x$ **b** $\frac{1}{5}$ **c** 13 **d** $\frac{x^2}{5}$

27 For each of the following expressions:

i Identify a pair of like terms.

ii Simplify the expression by combining these like terms.

a $4y + 9 + 8x + 5$

b $6x + 4 - 5x^2 + 7x$

c $4a + 2b - 6a - 3ab$

d $4xy - 4y^3 + 9x^3 - 4 - 7y^3 + 9x^3y$

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28 Determine whether the following statements represent a correct definition for a polynomial.

- **a** A polynomial is the sum of terms of the form ax^n, where a is any real number and n is any real number greater than or equal to zero.
- **b** A polynomial is the sum of terms of the form ax^n, where a is any whole number and n is any whole number.
- **c** A polynomial is the sum of a finite number of terms of the form ax^n, where a is any real number and n is a whole number greater than or equal to zero.
- **d** A polynomial is the sum of a finite number of terms of the form ax^n, where a is a whole number and n is any real number.

29 Create a binomial expression with a degree of 4 and a leading coefficient of 2.

30 Create a monomial expression with a degree of 0.

31 Fill in the blank with a single term that satisfies the given condition
$4x^3 - 5x + \square$

- **a** Creates a trinomial with a leading coefficient of 5
- **b** Creates a binomial with a leading coefficient of 2
- **c** Creates a trinomial with a degree of 3
- **d** Creates a polynomial with a degree of 1

32 Without repeating any values, fill in the blanks using only single-digit numbers to create a polynomial
$\square x^{\square} + \square x^{\square} - \square x^{\square} + \square x + \square$

- **a** With the lowest possible degree.
- **b** With the highest possible leading coefficient.
- **c** With the minimum number of terms.

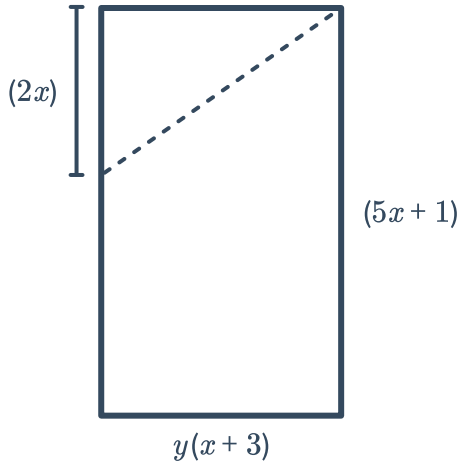
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33 A rectangle with the given dimensions is to have a right triangle cut out from one corner. Write and simplify a polynomial sum or difference to model each of the following:

a An expression for the length represented by y.

b An expression for the perimeter of the rectangle before the triangle has been removed.

c If the area of the rectangle is $5x^2 + 16x + 3$ and the area of the triangle is $x^2 + 3x$, determine the area leftover once the triangle has been removed.



34 Explain how two trinomials can be added together to produce a binomial.

35 A polynomial of degree m and a polynomial of degree n, with $m \geq n$, are added together. What is the highest possible degree of the result?

36 Fill in the blanks to make each equation true.

a $(x^3 - x + 7) + (\square x^3 + \square x + 3 = 10$

b $(\square x^2 - 5x - 5) - (\square x + 3) = 4x^2 - 2x - 2$

c $(\square x^{\square}) + (2x^3 - 3x + 1) = 2x^3 + x + 1$

d $(4x + 1) - (\square x^{\square} + \square) = 0$

37 A piece of paper with dimensions 8.5 inches by 11 inches will have a square with length x cut out from each corner to form a box.



Write and simplify a polynomial sum that can be used to represent the perimeter of the shape formed once the corners have been removed.

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38 Determine whether each statement is always, sometimes, or never true. Explain your reasoning with examples.

a Two polynomials added together will result in a polynomial.

b A linear function is a polynomial.

c A polynomial added to a non-polynomial will result in a polynomial.

d A non-polynomial added to a non-polynomial will result in a polynomial.

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